

Julia Nonnenkamp

Computer Science Ph.D. Student at UW–Madison interested in everyday cybersecurity and prevention of digital abuse & surveillance.

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Education

- 2024 – Present **University of Wisconsin–Madison** – Madison, WI
Ph.D. in Computer Science (*Focus: security & privacy*)
Advisor: Dr. Rahul Chatterjee
- 2020 – 2024 **Georgetown University** – Washington, DC
B.A. in Computer Science & Math
Mentors: Dr. Elissa Redmiles, Dr. Lisa Singh

Research experience

- July 2024 – Present **Graduate Student, UW–Madison**
MadS&P Research Group
Advisor: Dr. Rahul Chatterjee
Research Assistant (Spring 2024 onwards), Independent Study (Fall 2024)
- Demonstrated the feasibility of using GDPR/CCPA data exports to identify past or ongoing account compromise specifically within interpersonal threat models such as intimate partner violence. Develop safe and verifiable digital tools for survivors of technology-facilitated abuse using data engineering and trauma-informed interface design.
 - Used mixed methods to understand how adults with intellectual and developmental disabilities (IDD) interact with everyday security and privacy notifications.
- August 2023 – May 2024 **Undergraduate Researcher, Georgetown University**
Massive Data Institute & Georgetown Law
Advisors: Dr. Elissa Redmiles, Dr. Lisa Singh
Associated with Fritz Family Fellowship
- Examined law enforcement materials about software systems, algorithms, and verification procedures utilized in forensic DNA analysis. Built synthetic data generation pipelines (Python) to audit security risks for forensic DNA analysis software systems.

Peer-reviewed publications

- October 2025 **Hidden in Plain Bytes: Investigating Interpersonal Account Compromise with Data Exports.**
Julia Nonnenkamp, Naman Gupta, Abhimanyu Dev Gupta, Rahul Chatterjee.
ACM Computer and Communications Security (CCS). October 2025.
Top tier conference, 14% acceptance rate.

Talks & presentations

- February 2026 Flash Talk: The Madison Tech Clinic
Tech for Good Conference at the University of Chicago
- October 2025 Talk: Investigating Interpersonal Account Compromise with Data Exports
ACM Computer and Communications Security (CCS) Conference
- April 2024 Talk: The Technology behind Forensic Analysis
Fritz Family Fellows Conference
- April 2024 Poster: Auditing Probabilistic Genotyping Software used in Forensic DNA Analysis
Massive Data Institute Research Showcase
- December 2023 Poster: Risk Mapping in Forensic DNA Analysis
Massive Data Institute Research Showcase

Fellowships and grants

- 2026 Finalist, Aspen Institute Science and Technology Policy Fellowship
- 2025 Student Conference Travel Grant (ACM CCS)
- 2023 – 2024 Fritz Family Fellowship (Georgetown University)
Interdisciplinary research fellowship through GU's [Tech & Society Initiative](#).

Related work experience

- Summer 2024 **Two Six Laboratories — Research Intern** – Arlington, VA
Applied hybrid program analysis techniques to identify anomalous behavior in mobile messaging applications using Frida, Python, Haskell, and SMT solvers.
- June 2023 – May 2024 **U.S. Office of Management & Budget – Intern** – Washington, DC
Evaluated experimental applicability of transformer-based architectures for time-series forecasting of budget data. Refactored, tested, and documented econometric models in Python to improve code readability. Produced accessible, interactive visualizations for historical Presidential Budget data in JS and D3.

- 2022 – 2024 **Students of Georgetown, Inc. (The Corp)** – Washington, DC
- *Director of Data Analytics (2023-24)*: Built databases and ETL pipelines for point-of-sale data (PostgreSQL, Mage.ai, Python). Produced data communications and visualizations for store leads.
 - *Director of Operations, People Operations (2022-23)*: Ran payroll for 300+ employees, led Federal Work-Study partnership, managed company-wide onboarding.

Teaching experience

- Fall 2024 **Teaching assistant, CS 220: Data Science Programming** – UW–Madison
- Fall 2022 **Student academic assistant, COSC 160: Data Structures (C++)** – Georgetown
- 2021 – 2022 **Private Calculus (I-III) Tutor** – Washington, DC
- 2020 – 2021 **Instructor, Mathnasium** – Arlington, VA

Service & community leadership

- 2025 – Present **Mad S&P Seminar – Co-leader**
- 2024 – Present **Madison Tech Clinic – Technology Consultant, Volunteer Coordinator**
MTC is a volunteer partnership between UW–Madison and local intimate partner violence (IPV) advocacy groups and shelters. Through one-on-one consultations, we provide trauma-informed cybersecurity assistance to IPV survivors, helping them identify points of compromise and strengthen their security practices.
- 2021 – 2024 **Georgetown University Farmers Market – Board of Directors, Treasurer**
- 2022 – 2023 **Gaming Together – Volunteer Web Developer**
Maintained Gaming Together’s website (Bootstrap, PHP, XAMPP), implementing WCAG accessibility guidelines.

Technical skills

Reverse Engineering & Verification

Frida, Haskell, SMT solvers (Z3), Ghidra (novice), Rocq/Coq (novice).

Data Engineering & Analysis

Python (pandas/numpy, PyTorch), SQL, Mage.ai, Google Timesketch, OpenSearch (novice).

General

C++, Java, Flask, JavaScript (D3.js, Vue.js).